Attention as Investment: Algorithmic Management of Long Tail Creators on Short Video Platforms

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Abstract

To maximize short-term viewer engagement, online short video platforms like TikTok often prioritize the established, star creators in content exposure. However, creators in the long tail, who constitute the majority of creators with fewer followers and represent the platform's long-term future talent pipeline, have received limited attention in research and practice. Traditional long tail theory, developed in retail settings, characterizes long tail offerings as 'benign inventory' that expands surplus at low marginal cost. In contrast, in short video platforms, exposure of content from long tail creators consumes limited viewer attention and potentially displaces alternative viewing options, which raises uncertainty about how to manage long tail creators in these markets. In collaboration with two leading online short video sharing platforms in Asia, we report three complementary studies to investigate: i) the viewer engagement value (risk) of long tail creators' content to platforms and ii) the efficacy of viewer attention allocation, via algorithmic management, in cultivating long tail creators' development and growth. Study 1 reports a randomized field experiment removing long tail content from the recommender system. Contrary to the long tail theory's predictions, we find that, on average, long tail creators' content incurs a significant viewer engagement cost to the platform. Further heterogeneity analysis reveals that active viewers are more receptive to long tail content than casual viewers, indicating that long tail content's engagement value depends critically on viewer composition. Study 2 presents a second randomized field experiment on allocating additional viewer attention to long tail creators. Our results show that elevated viewer attention leads to significant increases in long tail creators' productivity, performance, and follower growth during the experimental window. Building on these findings, Study 3 reports a quasi-natural experiment that leverages exogenous variation in viewer activeness among viewer attention allocation to examine its short-term and long-term effects. We show that active (vs. casual) viewers lead to significant and positive increases in long tail creators' productivity (video uploads, video length, production days), performance (likes, comments, forwards, complete views), follower growth, and more importantly, monetization probability in the short and long runs. Further mediation analyses reveal engagement-based mechanisms: active viewers generate higher levels of engagement performance that stimulate long tail creators' long-term productivity, growth, and monetization outcomes. Our findings advance long tail theory by challenging the traditional theoretical consideration of long tail inventory holding and extending this theory into online content platforms. Furthermore, we contribute to research and practice on digital content creation and algorithmic management by demonstrating how platforms can effectively leverage viewer attention allocation in cultivating long tail creator development and growth.

Keywords: Long Tail Theory, Long Tail Creators, Digital Content Creation, Algorithmic Management, Viewer Attention Allocation, Viewer Engagement, Creator Development, Short Video Platforms

"Every artist was first an amateur."

– Emerson, R. W. (1870)

1 Introduction

Online short video platforms, for example TikTok,[1] YouTube Shorts,[2] and Instagram Reels[3] have experienced explosive economic expansion, with the global market estimated at $45.34 billion in 2024 and projected to reach $835.69 billion by 2035 (Chaudhary, 2025). These platforms serve as part of the digital infrastructure of 'the creator economy' (Bhargava, 2022), enabling and empowering the emergence of professions like content creators and influencers, who garner sizable income through monetization channels such as viewer subscriptions, livestream gifting, brand partnerships, and advertising revenue (Rieder et al., 2023; Archer et al., 2025).

Yet short video platforms face a significant power imbalance across the creator distribution. At the head of the distribution, a small number of star creators command the lion's share of viewer attention, engagement, and platform revenue (Han & Ravid, 2020; Maclean, 2021). Such concentration of power creates platform dependence on stars, prompting platforms to secure them through exclusivity agreements (Carroni et al., 2024). Meanwhile, star departures can damage platform performance (Forderer et al., 2025), exposing platforms to risks from creator turnover (Qian & Xie, 2025).

In contrast, creators in the long tail of the distribution on short video platforms receive limited attention from researchers and practitioners alike. Following the literature, we define 'long tail creators' as those with fewer than 1,000 followers, a threshold that captures creators who lack established audiences yet collectively comprise the vast majority of platform creators (Wu et al., 2023). Long tail creators constitute an important, largely underutilized, asset for short video platforms. First, they expand content diversity and innovation on platforms, enriching the overall content ecosystem (Kumar et al., 2014). Second, cultivating a robust long tail can reduce platforms' dependence on star creators, thereby

[1] https://www.tiktok.com/en/
[2] https://www.youtube.com/shorts/
[3] https://about.instagram.com/features/reels

mitigating risks from star creator bargaining power and possible exit (Cutolo & Kenney, 2021; Carroni et al., 2024). Third, long tail creators serve as the talent pipeline from which future stars emerge, providing platforms with a sustainable source of potentially high-value internal creators (Gaenssle & Budzinski, 2021). At the same time, long tail creators face severe barriers to discovery and growth, known as the cold-start problem (Lika et al., 2014; Liang et al., 2025): Although recommender systems nowadays seek to balance exploitation and exploration, they heavily rely on historical engagement when ranking and allocating viewer attention (Goldenberg et al., 2012). Long tail creators, lacking sufficient performance records, struggle to receive initial algorithmic exposure, which in turn prevents them from generating the engagement data needed to attract future attention allocation.

Drawing on the wisdom of the long tail theory traditionally developed in the retail contexts (Anderson, 2006), the long tail of product offerings constitutes 'benign inventory' with negligible marginal costs, which can generate consumer surplus through improved matching (Brynjolfsson et al., 2003, 2010). This logic suggests retail platforms should increase long tail exposure (Ratchford et al., 2022). Yet, such prediction may not be directly applicable to short video platforms. Particularly, viewer attention operates as a finite resource on those content platforms, allocated through recommender systems, creating direct competition among creators. Each viewer attention allocation necessarily displaces alternative content, whether from star creators or other long tail creators.

In turn, short video platforms face a central trade-off in managing long tail creators: Cultivating long tail creators may support content diversity and strengthen platform resilience in the long run, yet allocating viewer attention to long tail content may impose opportunity costs to viewers, and also risk engagement costs upon mismatch in the short run. Bearing the above in mind, this study empirically investigates how short video platforms can optimally manage their long tail creators, addressing the following set of research questions: *I) What is the viewer engagement value (risk) of algorithmically allocating viewer attention to long tail creators on short video platforms? II) How does algorithmic attention allocation influence short-term and long-term long tail creator outcomes (i.e., productivity,*

performance, and growth)? III) What could be the plausible underlying mechanisms of algorithmic attention allocation's effects on long tail creator outcomes?

To answer our research questions, we report three complementary studies in partnership with two leading short video platforms in Asia. Study 1 reports a randomized field experiment at the viewer-level, in which the platform removed long tail creators' content from the recommender system's feeds for the treatment group (*vs.* no such intervention in the control). We find that de-allocating long tail content significantly increases viewer engagement (e.g., watching time, number of likes, and number of complete views) in the treatment (*vs.* control) group, suggesting that long tail creators' content imposes short-term engagement costs to average viewers on the platform. Meanwhile, additional heterogeneity analysis reveals that the engagement value of long tail creators' content varies by viewer activeness. Specifically, the engagement cost of long tail content is smaller for active viewers than for casual viewers, indicating that active viewers are more receptive to long tail content.

Study 2 presents another randomized field experiment at the creator-level, examining whether the treatment of algorithmically allocating additional viewer attention towards long tail creators (*vs.* no such intervention in the control) facilitates their development during the experimental period. Our results show that elevated viewer attention in the treatment (*vs.* control) group significantly improves long tail creators' productivity, reflected in the number of videos uploaded, video length, and days of production, as well as their content performance, including the number of likes, forwards, comments, and complete views. The increased viewer attention also translates to long tail creators' growth, as measured by number of new followers. These findings suggest that while long tail content imposes short-term engagement costs on viewers, algorithmic allocation of viewer attention can effectively promote long tail creator development and growth on the platform.

Building on Studies 1 and 2, Study 3 investigates how the composition of viewer attention, specifically those from active viewers, impacts long tail creators' development in both the short and long run. Through a quasi-natural experiment, we leverage exogenous variation in viewer activeness among creators' initial viewer attention allocation during their first 30 days after account registration. Our results

show that greater early exposure to active viewers leads to significant and persistent improvements in long tail creators' productivity, content performance, and growth, measured over both a short-term one-month window and a long-term six-month horizon. Notably, we find that higher early active-viewer attention increases long tail creators' monetization probability in both the short and long terms. Our further mediation analyses provide evidence for engagement-based mechanisms: Active viewers' initial attention generates stronger engagement performance for creators (e.g., likes, comments, forwards, and complete views), which in turn facilitates long tail creators' subsequent productivity, follower growth, and monetization outcomes. Taken together, these findings suggest that algorithmic allocation of viewer attention can function as a form of forward-looking investment in the development of long tail creators on short video platforms.

This work seeks to contribute to the related literature on long tail theory, digital content creation, and algorithmic management. First, this research extends the long tail theory from retail contexts to short video platforms, challenging the traditional consideration of long tail products constituting 'benign inventory' in retail settings (Anderson, 2006; Brynjolfsson et al., 2003). Specifically, our empirical findings reveal that long tail creators' content incurs short-term engagement costs on average viewers on short video platforms, contrasting with previous predictions of long tail products in retail. Additionally, unlike retail products that remain passively displayed on the shelf (Ratchford et al., 2022), our findings show that long tail creators actively evolve, given viewer attention. Over time, such attention enables long tail creators' capability development, growth, and monetization probability, pointing to the potential of cultivating a sustainable creator ecosystem.

Second, our work contributes to the digital content creation literature by advancing our understanding of the vast but understudied long tail creators, shifting attention from the stars (Forderer et al., 2025; Qian & Xie, 2025). Prior research has examined various platform interventions to support user content creation, including monetary incentives such as revenue-sharing (Jain & Qian, 2021) and non-monetary mechanisms like symbolic awards or performance feedback (Burtch et al., 2022; Gallus, 2017; Huang et al., 2019). However, these interventions often assume creators already receive certain visibility

and engagement to be noticed and selected for such programs. In the current creator economy, long tail creators, especially later entrants on maturing platforms, face cold-start problems in accessing monetary incentives tied to performance thresholds. We demonstrate that viewer attention allocation through algorithmic intervention can address this challenge. Importantly, we examine both short-term and long-term developmental outcomes, showing that the benefits of attention allocation persist beyond the initial intervention period. We also explore the plausible mechanisms through which active viewership drives sustained long tail creator development through content engagement signals.

Third, this study advances the literature on algorithmic management (Kellogg et al., 2020; Duggan et al., 2023) by examining how platforms manage algorithms in navigating the tension between optimizing short-term viewer engagement and cultivating long-term ecosystems. In general, algorithms tend to favor established creators who generate predictable engagement, while neglecting the long tail. We demonstrate that algorithmic attention allocation offers the scalable intervention to address this challenge. Unlike monetary incentives or manual curation that require significant resources and creator selection (Burtch et al., 2018), algorithmic attention allocation can systematically reach long tail creators who would otherwise remain invisible. Further, our findings suggest that the effectiveness of algorithmic management depends on the composition of allocated attention: Active viewers generate stronger content engagement that stimulates long tail creators' long-term development and growth.

Our findings also provide practical implications for short video platform practitioners. For platform managers and designers, our results show that long tail creators are not 'free inventory.' Long tail content exposure competes for finite viewer attention and can impose short-term engagement costs for average viewers. Managers should therefore explicitly manage viewer attention via algorithmic allocation, balancing current engagement metrics with the need to cultivate a pipeline of future successful creators. In addition, our results point to the importance of viewer composition, as exposure to active viewers yields better long tail creator gains while somewhat mitigating engagement losses. Designers can use this insight to improve recommender systems, as well as creator growth and incubation programs. For creators, especially those in the long tail, our results show that creators can meaningfully grow over time,

and they benefit more from exposure that generates active engagement than from passive views. Creators, therefore, want to prioritize engagement-oriented content strategies, especially during early exposure windows of algorithmic learning and follower base formation.

2 Theoretical Foundation & Related Literature

2.1 The Long Tail Theory

The long tail theory posits that, as technological advancements alleviate the physical constraints of product distribution and search costs, consumer demand would shift from a small number of blockbuster 'hits' to the vast catalog of other products on the market (Anderson, 2006). Shifting away from the traditional '80/20' Pareto principle on consumption patterns that roughly 80 percent of revenue is generated by the 20 percent top-selling products (Pareto, 1964), the long tail theory considers the increasing importance of the large number of 'tail' products, as this expanded access to mass products has been shown to generate considerable consumer surplus (Brynjolfsson et al., 2003, 2010). Prior research has also demonstrated that online channels tend to exhibit significantly less concentrated sales distributions than offline channels, due to broader product offerings and lower consumer search frictions (Brynjolfsson et al., 2003, 2011).

Extant work on the long tail theory has explored factors that influence consumer demand for long tail offerings. For example, previous studies have shown that low-cost information discovery, greater product accessibility through online channels, and firm promotion strategies can expand demand toward the long tail (Kumar et al., 2014; Zentner et al., 2013; Zhang, 2018). Conversely, other forces have been shown to increase demand concentration toward the top. For instance, research has found that biases in online word-of-mouth, design features of mobile channels, and behavioral scenarios like choice overload or herding under uncertainty can hinder consumer exploration and lead to more concentrated consumption patterns (Gu et al., 2013; Park et al., 2020; Aparicio et al., 2025; Geva et al., 2024).

Prior research has primarily considered the 'long tail products' in retail settings (Brynjolfsson et al., 2010; Ratchford et al., 2022), where the large number of relatively low-demand products constitute the 'benign inventory.' In online retail, digital shelf space and listing costs are marginal, so long tail stock

keeping units (SKUs) do not directly displace existing offerings. Non-consumption of these long tail products imposes, at most, negligible cost on the retailer, and therefore the long tail is treated as inventory that is 'safe to carry.' Meanwhile, on the demand side, discovery of long tail products is predominantly consumer pull-driven. That is, consumers initiate search, filtering, and browsing when they have specific or niche preferences, and recommendation and search tools mainly reduce the frictions of finding items that already fit their tastes. Therefore, carrying and expanding the offerings of long tail products, combined with reduced search costs and distribution frictions, generally benefit consumers as well as the retail platform.

However, content platforms operate under fundamentally different conditions. In this context, the 'long tail creators' require additional considerations. First, discovery of long tail content/creators is primarily push-driven through algorithmic feeds rather than pull-driven consumer search (Felaco, 2025). Long tail creators are injected into finite, scarce viewer attention slots, so every exposure to long tail content directly displaces some alternative, often a head creator with predictable engagement (Goldhaber, 2006). This is highly risky to the content platform, because when recommended long tail content is a poor fit, it does not generate zero demand as in retail; it can even reduce engagement and weaken users' interest in the platform. Thus, in contrast to the 'benign inventory' in retail, long tail content/creators in content platforms are not always 'safe to carry' at the point of exposure, given the potential misfit and engagement cost.

Second, long tail creators are strategic, adaptive individuals rather than passive SKUs (Duffy & Meisner, 2023). The long tail creators continuously observe performance feedback, such as impressions, click-throughs, watch time, follower growth, and update their behaviors in response. Given that long tail creators' productivity and content quality can improve with experience and guidance, content platforms do not just 'list' those creators; instead, long tail creators can be investment opportunities for the content platform through approaches like exposure, analytics, training, and monetization opportunities. Third, unlike retail platforms that realize value through discrete purchases, the value creation for content platforms depends on sustained engagement between creators and viewers over time (Engert et al., 2023).

Content platforms' success is thus tied less to any single 'sale' and more to fostering a long-term creator-viewer relationship that supports sustained engagement and future monetization.

Bearing the above in mind, managing 'long tail creators' in content platforms requires delicate considerations of viewer engagement, algorithmic allocation, and sustained participation. Content platforms face a central trade-off with long tail creators: On one hand, the platform can maintain exposure of popular creators that appeal to broad audience tastes, which ensures viewer engagement but creates dependency on a small number of 'superstars,' whose departure can threaten platform stability (Forderer et al., 2025). On the other hand, the platform needs to expand exposure to long tail creators, which broadens discovery and improves viewer taste learning, yet may dilute attention away from head creators who disproportionately drive user traffic and aggregate consumption (Qian & Xie, 2025). This trade-off reflects an important tension between short-term engagement optimization and long-term ecosystem development. Our study seeks to empirically investigate the value of long tail creators on content platforms by examining how viewer engagement responds to exposure of their content and whether and how such exposure translates into persistent creator growth.

2.2 Digital Content Creation

Enabled by platform technologies, digital content creation encompasses formats such as online word-of-mouth, knowledge sharing, audios, and videos (Bapna et al., 2018; Zhang & Zhu, 2011; Schanke et al., 2024; Zhao et al., 2025). Early work explored the consequences of digital content (Duan et al., 2008; Zhu & Zhang, 2010), and later studies examined the antecedents of digital content creation, including creator characteristics and broader external factors. Examples of creator characteristics include psychological distances (Huang et al., 2016), the pursuit of social status and reputation (Goes et al., 2016; Fu et al., 2018), and strategic content differentiation (Levina & Arriaga, 2014). Meanwhile, various external factors have also been shown to influence digital content creation, including market competition (Wu & Zhu, 2022), platform design choices like recommender systems (Qian & Jain, 2024), content feed integration (Cao et al., 2024), content length policies (Gu & Zhao, 2024), and advertising models (Ren, 2025; Zhang et al., 2025).

Prior work typically treated creators as a relatively homogeneous group, with a few exceptions looking at 'star'/highly popular creators (Forderer et al., 2025; Qian & Xie, 2025). Existing studies show that star creators generate disproportionate value for the platforms (Han & Ravid, 2020; Maclean, 2021), platforms use exclusivity arrangements to retain them (Carroni et al., 2024), and the exit of these top creators can inflict significant negative shocks on platform performance (Forderer et al., 2025). Thus, over-reliance on star creators can lead to strategic vulnerability, exposing platforms to risks from creator turnover and power imbalance (Bhargava, 2022).

Meanwhile, much less is known about the vast 'long tail creators' who collectively comprise the majority of the creator population but individually receive limited attention (Wu et al., 2023). Long tail creators often face the 'cold-start problem' (Lika et al., 2014; Liang et al., 2025) on content platforms: Recommendation algorithms rely on historical data to predict relevance and maximize viewer engagement, but for long tail creators with limited performance history, recommendation algorithms face greater prediction uncertainty. In turn, the content from long tail creators is less likely to be picked up by recommendation algorithms, which then generates weak engagement and further suppresses future visibility. This self-reinforcing loop creates higher barriers to discovery, slower early growth, and greater risk of creator disengagement and exit over time.

However, long tail creators are valuable to content platforms for several reasons. First, they contribute to content variety on the platform, thereby expanding the platform's breadth of content offering (Kumar et al., 2014). Second, long tail creators constitute the platforms' talent pipeline, as successful creators may start their trajectories in the long tail (Gaenssle & Budzinski, 2021). Third, cultivating a robust long tail base can reduce platforms' dependence on a handful of star creators, strengthening resilience against creator exit or bargaining power shifts (Cutolo & Kenney, 2021; Carroni et al., 2024). Accordingly, it is important to advance our knowledge on how content platforms can help develop and foster the growth of long tail creators.

Platforms generally use monetary incentives and/or non-monetary interventions to cultivate creator development. For example, platforms can compensate creators through different business models,

such as revenue-sharing from advertising (Jain & Qian, 2021), or by allowing monetization through subscriber-only content (Li et al., 2024) or affiliate marketing (Gu et al., 2025). Additionally, platforms can deploy non-monetary interventions, such as symbolic awards (Gallus, 2017; Burtch et al., 2022), performance feedback (Huang et al., 2019), or social nudges (Zeng et al., 2023).

Yet, nurturing and developing long tail creators poses a unique challenge to content platforms. Unlike star creators who bring in measurable financial resources (Qian & Xie, 2025), long tail creators mostly generate little or no revenue, and thus monetary incentives to long tail creators are financially unviable at scale. These considerations point platforms toward non-monetary interventions. Some of the non-monetary mechanisms studied in prior work, such as badges, symbolic recognition, peer comparisons, or performance dashboards, may help, but their effectiveness hinges on realized, recognizable performance records (Kuhnen & Tymula, 2012; Huang et al., 2019).

It is notable that for long tail creators, the core challenge is that their content is rarely discovered in the first place, leaving them with limited opportunities to be observed, evaluated, and appreciated. In this study, we extend related prior work by focusing on the intervention of 'viewer attention allocation' that precedes performance feedback. Viewer attention allocation involves the content platform deciding how much and what type of viewer attention to direct to long tail creators via algorithmic recommendations. We posit that viewer attention is a finite, platform-allocatable resource that incurs engagement costs, and through its volume and composition, can potentially influence whether and how long tail creators can meaningfully develop and grow, both short-term and long-term.

2.3 Algorithmic Management

This study also connects to the body of work on algorithmic management, particularly the notion of platforms using algorithms to perform core managerial tasks such as recommendations, resource allocation, monitoring, and evaluation (Kellogg et al., 2020; Duggan et al., 2023). For example, in a systematic review of research on algorithm management in the gig economy, Kadolkar et al. (2025) consider algorithm management as a multi-dimensional system of algorithmic control and monitoring,

deliberately created information and power asymmetries, and dynamic pricing strategies, whose combined effects influence human participants' working experience and performance.

Our work centers on recommendation algorithms, which play an important role on content platforms (Bhargava, 2022). Previous research has demonstrated the efficacy of recommendation algorithms in enhancing users' content consumption and engagement. For instance, Zhou et al. (2010) find that YouTube's recommender system increases the diversity of video views. Claussen et al. (2019) observe higher user clicks with personalized recommendation algorithms compared to human-curated ones in online news. Additionally, Holtz et al. (2020) report a significant increase in podcast streams after switching from popularity-based to recommendation algorithms.

At the same time, some scholars have explored unintended consequences of recommendation algorithms, such as Lambrecht and Tucker (2019), who find gender-neutral 'science, technology, engineering, and mathematics' (STEM) career ads are shown more to men due to algorithms optimizing for cost-effectiveness on Facebook. Other work, like Brown et al. (2022), discusses how algorithms contribute to polarization and 'echo chambers' by recommending content that aligns with users' existing beliefs. Also, Anderson et al. (2020) report a field experiment on Spotify, showing that recommendation algorithms reinforced users' existing preferences, reducing content diversity and exploration.

In response, a growing body of research focuses on the technical development of recommendation algorithms towards management objectives. For instance, Kim et al. (2019) propose a sequential and diverse recommendation (S-DIV) algorithm on blog platforms, designed to learn users' temporal preferences and jointly optimize accuracy and diversity. Chen et al. (2024) introduce a deep learning approach for recommending background music for short videos, using user-music and video-music matching with attention-based aggregation.

Besides technical work, other researchers employ analytical approaches, seeking to optimize factors related to existing recommender systems to better serve the interests of different stakeholders, such as platforms, content creators, and users, without altering the underlying algorithms. For example, Qian and Jain (2024) develop a game-theoretic model to study how recommender systems and revenue-

sharing plans impact content creation, profits, and welfare on online short video platforms. They find that biasing recommendations could incentivize creators to invest in quality, resulting in a win-win scenario for both the platform and creators.

This work extends the related empirical investigations that examine algorithm management on digital platforms (Goldenberg et al., 2012; Hosanagar et al., 2014). One closely relevant paper is Zeng et al. (2024), which examines how excluding highly popular creators from recommendations affects users' dual roles as consumers and producers. Our study differs in several key aspects. First, rather than considering the 'stars,' we focus on long tail creators, a strategically important yet under-studied creator group. We analyze the treatment of 'viewer attention allocation' as a form of algorithmic control that manages these creators' opportunities to accumulate performance records on the content platform. Second, instead of focusing only on production quantity during a short observational window, we employ both short-term and long-term lenses, examining developmental outcomes like productivity, performance, follower growth, and monetization potential. Third, we seek to unpack the mechanisms through which viewer attention allocation operates, elucidating how exposure and the composition of allocated viewership shape performance and, in turn, creator growth. In doing so, our study speaks to the efficacy of algorithmic management for cultivating and sustaining long tail creator development and growth.

3 Research Framework & Study Context

3.1 Research Framework

Building on the above literature, this work consists of a series of three studies, aiming to provide a comprehensive understanding of the value (and risk) of allocating viewer attention to long tail creators, the creator-side developmental returns to viewer attention allocation, and the mechanisms through which attention allocation possibly helps cultivate long tail creator growth. To begin with, we report a randomized field experiment that examines the viewer engagement variation from de-allocating attention to long tail creators on the digital platform (Study 1). We also explore the heterogeneity among viewers, particularly how viewer activeness moderates the extent to which viewers value long tail creators and their content, as manifested in viewer engagement. Next, through another randomized field experiment,

we report the efficacy of viewer attention allocation in motivating long tail creators' productivity, performance, and follower growth during the experimental period (Study 2). Furthermore, by analyzing archival data in a quasi-experimental setting, we explore how active viewership among viewer attention allocation influences long tail creator outcomes, including short-term as well as long-term productivity, performance, and growth (Study 3). Our overarching research framework is depicted in Figure 1.

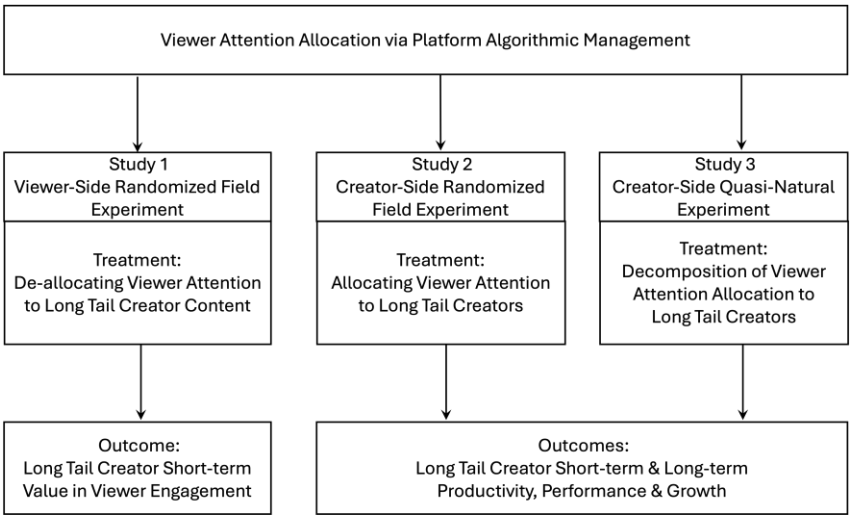

Figure 1. Research Framework

3.2 Study Context

Our study context is set in two leading online short video platforms in Asia, hereafter referred to as Partner Platforms A and B.[4] Both platforms have substantial user bases, with multi-millions in daily active users (DAUs). Functionally, they closely resemble TikTok, primarily featuring short videos ranging from a few seconds to a few minutes in length. Both platforms operate highly comparable user interfaces (UIs) centered around a 'recommendation page,' as shown in Figure 2.[5] This page displays a seamless, scrolling feed of videos continuously recommended to individual viewers based on their interests, watch history, and engagement behaviors. Both platforms employ industry-standard recommendation algorithms and systems comparable to those used by TikTok and other major short video sharing platforms.[6] As two-

[4] The names of the partner platforms are anonymized per confidentiality agreements.

[5] Figure 2 presents a screenshot of TikTok's user interface (UI) with key features labeled. The UI features of both partner platforms closely resemble this design.

[6] Appendix A explains typical recommender systems used in short video sharing platforms.

sided markets, the platforms connect content creators with viewers. Creators aim to attract viewership, engagement, as well as followers, and are financially incentivized through revenue-sharing arrangements based on performance metrics. Viewers access videos on the platforms free of charge and consume a vast array of content across verticals such as entertainment, education, lifestyle, and sports. The content platforms generate revenue primarily through advertising and sponsorships, a portion of which is shared with creators to encourage sustained content production.

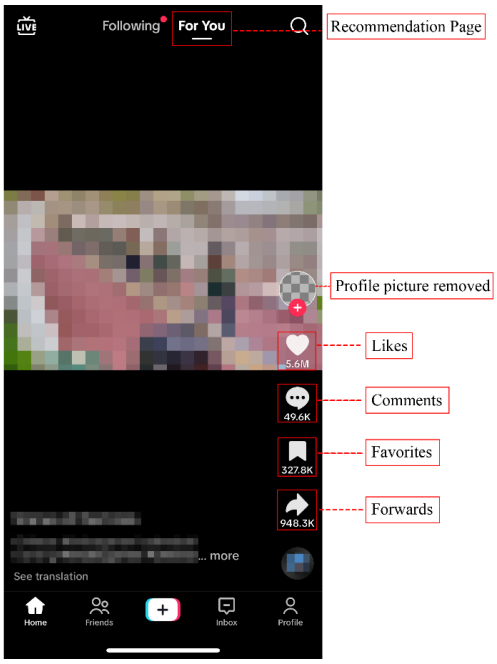

Figure 2: Screenshot of the User Interface on TikTok

4 Methods & Results

4.1 Study 1: Viewer-Side Randomized Field Experiment

4.1.1 Experimental Design in Study 1

To investigate the value of long tail creators' content to viewer engagement, Partner Platform A conducted a viewer-side randomized field experiment on December 2-12, 2022, employing a between-subjects design. Specifically, viewers were randomly assigned into two groups: A treatment group, where 50% of the videos produced by long tail creators were blocked from the candidate video pool feeding into the recommendation algorithm for each viewer, and a control group without such intervention. In other

words, the treatment manipulation involves the partner platform randomly blocking content by a portion of the long tail creators, operationalized as creators with fewer than 1,000 followers at the time of the experiment,[7] from entering the candidate video pool that could have been recommended to treated viewers.[8] This approach ensured that the treatment intervention did not directly modify the recommender system itself but only manipulated the video supply from being distributed by the recommendation algorithm (Kellogg et al., 2020). We summarize the experimental design of Study 1 in Table 1.

Table 1: Experimental Design in Study 1

Group	***Manipulation***	***Observations***
Treatment	Block 50% of long tail creators' video supply from the candidate video pool of each viewer	4,532,431
Control	N/A	9,065,114

Notes: Observations correspond to unique viewers included in the experiment. Randomization is conducted at the viewer level.

4.1.2 Data and Measures in Study 1

We obtained viewer demographic and engagement data for all viewers who participated in the randomized field experiment in Study 1. Table 2 presents the definitions of key variables, and Table 3 reports their descriptive statistics. We also report the results of randomization checks and manipulation checks in Appendix B.

Table 2: Variable Descriptions in Study 1

Variable	***Description***
Independent Variable	
Treatment	A viewer's assignment to the experimental group (0 = Control, 1 = Treatment)
Viewer Demographics	
Gender	A gender indicator (0 = Unknown,1= Male, 2 = Female)
CityLevel	The tier of a viewer's located city (0 = Unknown, 1 = First-tier city, 2 = Second-tier city, 3 = Third-tier city, 4 = Fourth-tier city, 5 = Fifth-tier and over Fifth-tier city)
NumFriends	The total number of friends that a viewer has on the platform
NumFollowers	The total number of followers that a viewer has on the platform
Viewer Engagement	
WatchingTime	The total duration of video watched by a viewer during the experiment (in seconds)
NumComments	The total number of comments a viewer posted on videos during the experiment
NumForwards	The total number of videos a viewer forwarded to others during the experiment
NumLikes	The total number of likes a viewer left on videos during the experiment
NumCompleteViews	The number of video views in which the viewer watched at least 90% of the video's total duration during the experiment period

[7] This cutoff also aligns with common size-based classifications in the influencer/creator literature, where the smallest tier ("nano") of influencers is defined as having fewer than 1,000 followers (Chen et al., 2024).
[8] The content creators and viewers were unaware of this manipulation on the partner platform. And the treated users accounted for less than 1% of the platform's total traffic, ensuring minimal disruption to the overall ecosystem.

Others	
Active	A binary indicator of whether a viewer is classified as active by the platform (0 = Casual, 1 = Active)
NumViews	The total number of videos presented to the viewer in their feed during the experiment, regardless of the watching duration
LongTailViewShare	The proportion of video views that come from long tail creator content relative to the total video views by a viewer during the experiment

Table 3: Descriptive Statistics in Study 1

Variable	***Obs.***	***Mean***	***Std***	***Min***	***Max***
Viewer Demographics					
Gender	13,597,545	1.48	0.52	0	2
CityLevel	13,597,545	3.40	1.29	0	5
NumFriends	13,597,545	456.80	704.23	0	22,288
NumFollowers	13,597,545	0.40	6.24	0	4,787
Viewer Engagement					
WatchingTime	13,597,545	7,659.90	19,364.27	0	637,114
NumComments	13,597,545	0.22	3.45	0	550
NumForwards	13,597,545	1.00	6.95	0	550
NumLikes	13,597,545	2.24	13.47	0	550
NumCompleteViews	13,597,545	134.45	330.91	0	14,853
Others					
Active	13,597,545	0.04	0.20	0	1
NumViews	13,597,545	333.60	901.29	0	61,923
LongTailViewShare	13,597,545	0.10	0.12	0	1

Notes: (a) To protect data confidentiality, the partner platform scaled the raw values of several variables (*NumFriends, NumFollowers, WatchingTime, NumComments, NumForwards, NumLikes, NumCompleteViews, NumViews*). Therefore, the numerical values presented in the table represent scaled metrics rather than absolute values. (b) All variables are measured at the viewer level to investigate the effect of blocking long tail creators' content on viewer engagement. (c) *NumViews* and *LongTailViewShare* are used in the Manipulation Checks in Appendix B. (d) *NumComments, NumForwards,* and *NumLikes* are capped at platform-defined upper thresholds to exclude outliers and abnormal activity. (e) *NumViews* refers to the traffic or attention directed to the viewer, specifically how many videos are presented to the viewer in their recommendation feed. *NumCompleteViews* measures viewer engagement, specifically the number of videos a viewer has consumed at least 90% of during the observation period.

4.1.3 Estimation Equations in Study 1

Our experimental manipulation was at the viewer level, where we varied viewers' exposure to long tail creator content to assess its impact on viewer engagement outcomes. Thus, the unit of analysis centers on the viewer level. Specifically, to examine the main effects of de-allocating long tail creator content on viewer engagement, we estimate the following Equation (1.1) with ordinary least squares (OLS) regressions:

$$OutcomeVariable_v = \beta_0 + \beta_1 \times Treatment_v + \varepsilon_v \tag{1.1}$$

where $Treatment_v$ is a binary indicator for whether viewer v is in the treatment group (where 50% of long tail creators' videos were blocked from the recommendation pool) versus the control group.

$OutcomeVariable_v$ represents our engagement metrics of interest, measured at the viewer level during the experiment period. Our primary outcomes focus on overall viewer engagement on the platform, including total video watching time (*WatchingTime*), the number of comments (*NumComments*), forwards (*NumForwards*), likes (*NumLikes*), and complete views (*NumCompleteViews*).

Next, we explore the heterogeneity in the effects of long tail creators' content on viewer engagement by viewer activeness. Previous research indicates that online platform users differ in their activeness and exhibit different behavioral patterns accordingly (Heinonen, 2011). Adapting these prior considerations to the context of the online video sharing platform, we consider that active viewers, who frequently engage with the platform and consume a greater amount of content, may have different preferences and responses compared to casual viewers, who use the platform less frequently. Thus, we test how active viewers' response towards the treatment differs from that of casual viewers, estimating the following Equation (1.2):

$$OutcomeVariable_v = \beta_0 + \beta_1 \times Treatment_v + \beta_2 \times Active_v + \beta_3 \times Treatment_v \times Active_v + \varepsilon_v \quad (1.2)$$

where $Active_v$ is a binary variable indicating whether viewer v is classified as an active (*vs.* casual) viewer based on their engagement level one week prior to the experiment (November 25–December 1, 2022). Active viewers are defined as individuals whose number of video views in the week preceding the start of the experiment exceeds a certain threshold set by the platform.[9] The coefficient β_3 allows us to test whether active (*vs.* casual) viewers respond differently to changes in long tail creator content.

4.1.4 Results in Study 1

We first report the main results of Study 1 in Table 4. The viewers in the treatment group (where 50% of long tail creators' videos were blocked from the recommendation pool) significantly increased their number of likes ($\hat{\beta}_1 = 0.017$, $p < 0.05$), number of complete views ($\hat{\beta}_1 = 0.585$, $p < 0.01$), and total

[9] The partner platform did not share the exact raw value of the threshold due to data sensitivity. Note that among the overall platform, active users constitute about 1.80% of the viewer population, and they account for 49.50% of total viewership and 42.09% of total video watching time. In our experiment, active viewers represent approximately 4% of the sample.

video watching time ($\hat{\beta}_1 = 24.906, p < 0.05$), respectively, compared to the control group.[10] The effects on comments and forwards are statistically insignificant. Note that since our treatment removed long tail creators' content, the positive effects indicate that long tail content, on average, *negatively* influences viewer engagement.

Table 4: Main Effects of De-allocating Long Tail Creator Content on Viewer Engagement in Study 1

Variable	*NumLikes* (1)	*NumCompleteViews* (2)	*WatchingTime* (3)	*NumComments* (4)	*NumForwards* (5)
Treatment	0.017**	0.585***	24.906**	0.001	0.002
	(0.008)	(0.190)	(11.216)	(0.002)	(0.004)
Observations	13,597,545	13,597,545	13,597,545	13,597,545	13,597,545

Notes: (a) Partner Platform A scaled the raw values of outcome variables in Studies 1 and 2. Therefore, the coefficients presented in this and subsequent tables represent scaled metrics rather than absolute values. (b) Robust standard errors in parentheses; $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

Next, we present the heterogeneous treatment effects by viewer activeness in Table 5. As a sanity check, the direct effects of the active indicator ($\hat{\beta}_2$) are positive and significant across all outcomes, indicating that active viewers are substantially more engaged than casual viewers in general. Meanwhile, the interaction terms between treatment and active viewer status are negative and significant for number of complete views ($\hat{\beta}_3 = -2.480, p < 0.05$) and watching time ($\hat{\beta}_3 = -86.045, p < 0.05$), indicating that de-allocating long tail creator content results in less engagement for active viewers, and active (*vs.* casual) viewers have stronger preferences for content from long tail creators. Interestingly, even with less long tail content, active viewers maintained their baseline habit of commenting more frequently than casual viewers ($\hat{\beta}_3 = 0.051, p < 0.01$).

Table 5: Heterogeneous Treatment Effects by Viewer Activeness in Study 1

Variable	*NumLikes* (1)	*NumCompleteViews* (2)	*WatchingTime* (3)	*NumComments* (4)	*NumForwards* (5)
Treatment	0.019**	0.585***	24.691***	-0.002	0.003
	(0.008)	(0.190)	(8.831)	(0.002)	(0.004)
Active	10.020***	1200.000***	58,431.645***	1.831***	6.455***
	(0.021)	(0.991)	(24.597)	(0.005)	(0.011)
*Treatment*Active*	-0.049	-2.480**	-86.045**	0.051***	-0.008
	(0.037)	(1.037)	(42.588)	(0.009)	(0.019)
Observations	13,597,545	13,597,545	13,597,545	13,597,545	13,597,545

[10] It is worth noting that a 0.33% increase in video watching time can be quite meaningful for a large-scale short video platform. TikTok users are estimated to spend approximately 95 minutes per day on the app (DataProt, 2023). A 0.33% increase would translate to an additional 19 seconds of daily engagement per user. For a platform with hundreds of millions of daily active users, such as Partner Platform A, this seemingly small percentage increase amounts to millions of additional hours of total viewing time per day across the entire user base.

Notes: Robust standard errors in parentheses; $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

Further, we explore the underlying mechanisms of the results in Study 1, starting with interviewing the product managers and data scientists of Platform A. They commented that long tail creators typically produce more diverse and innovative content, catering to various viewer tastes, while popular star creators focus on the mainstream appeal. A product manager noted that, 'long tail creators are likely to publish innovative content that diverges from the platform's mainstream trends.' Another added, 'long tail creators often venture into niche areas, unlike popular creators who target mass appeal.' The head of the Data Analytics team observed that, 'the diversity of content from long tail creators likely surpasses that of other creators.' Based on the insights from those practitioners, several plausible mechanisms arise. First, the content of long tail creators might be perceived as low quality, explaining the positive effect of removing it on the average viewer engagement. Second, long tail creators are more likely to produce diverse content. Removing such content reduces consumption diversity, which may affect the active viewers more than the casual ones.

To test the content quality and diversity mechanisms, we obtained a dataset containing a random sample of 188,374,718 creators from Platform A, observing their characteristics and content production behaviors over a 7-day period. This dataset allows us to compare the attributes of long tail creators with those of other creators on the platform. For content quality, we use *CompleteViewRate*, defined as the proportion of video views that finish watching at least 90% of the video's total duration, as a proxy for perceived content quality. A two-tailed t-test comparing the *CompleteViewRate* of long tail creators to that of other creators indicates that long tail creators have a significantly lower *CompleteViewRate* (15.70%) than other creators (36.02%) ($p < 0.01$), suggesting that the content produced by long tail creators is of lower perceived quality. For content diversity, we follow prior literature and adopt the Herfindahl-Hirschman Index (HHI) of video topics as the key metric[11] (Cheng & Nault, 2012; Clark et al., 2023). The

[11] The HHI is calculated as follows: $HHI_c = \sum_{j=1}^{M} p_j^2$, where HHI_c indicates the Herfindahl-Hirschman Index (HHI) for a content creator c, M is the total number of unique video topics produced by creator c, and p_j represents the proportion of videos that creator c has produced on topic j, relative to their total video production.

HHI measures the concentration of videos produced by a creator across different topics, where a lower HHI value indicates greater content diversity and a higher HHI value suggests more topic-focused content production. A two-tailed t-test comparing the HHI of long tail creators to that of other creators shows that long tail creators have a significantly lower average HHI (0.10) than other creators (0.15) ($p < 0.01$), suggesting that long tail creators produce content covering a wider variety of topics and thus contribute to greater overall content diversity on the platform. Together, these findings provide supporting evidence for both the content quality and content diversity mechanisms.

To summarize, in Study 1, we find that on average, de-allocating long tail content to viewers increases overall engagement, which suggests that the content from long tail creators actually impedes viewer engagement on the platform. This result challenges and advances the traditional consideration of the long tail theory in retail settings that expanding product variety typically generates surplus. In online content platforms, long tail creators' content appears to be not 'safe to carry' in the same way as long tail products, as exposure to long tail content competes for finite viewer attention and can impose short-term engagement costs. Meanwhile, active viewers, who contribute disproportionately to platform engagement, are significantly less deterred by long tail creators' content and, to some extent, exhibit stronger engagement and preferences for it than casual viewers.

Taken together, our findings point to a central trade-off in online platform algorithmic management: Allocating viewer attention to long tail creators entails short-term engagement costs, yet unlike long tail products, creators are adaptive individuals whose productivity, performance, and growth can improve over time. This leads to the next question: Can viewer attention allocation function as a platform investment strategy that cultivates long tail creators' development and growth? Study 2 seeks to directly answer this question by examining whether allocating viewer attention to long tail creators stimulates their productivity, performance, and growth in a field experiment setting.

4.2 Study 2: Creator-Side Randomized Field Experiment

4.2.1 Experimental Design in Study 2

To examine how allocating viewer attention affects long tail creators' development and growth, Partner Platform A conducted a creator-side randomized field experiment from March 18 to April 1, 2022. The experiment employed a between-subjects design on a random sample of long tail creators, who were then randomly assigned into either treatment or control groups. In the treatment group, long tail creators received additional platform-allocated viewer attention to their content, thereby boosting their content exposure and viewership, through an enhanced weighting of their content in the recommender system. Conversely, long tail creators in the control group did not receive such an intervention. Table 6 shows the experimental design in Study 2.

Table 6: Experimental Design in Study 2

Group	***Manipulation***	***Observations***
Treatment	Allocate additional viewer attention to long tail creators	480,048
Control	N/A	480,353

Notes: Observations correspond to unique creators included in the experiment. Randomization is conducted at the creator level.

4.2.2 Data and Measures in Study 2

We obtained data from Partner Platform A to analyze the randomized field experiment in Study 2. The dataset includes creator-level information on demographics, content productivity, performance metrics, and follower growth within the experiment window. Table 7 presents the definitions of key variables in Study 2, and Table 8 shows the descriptive statistics of these variables. We report the results of randomization checks and manipulation checks in Appendix C.

Table 7: Variable Descriptions in Study 2

Variable	***Description***
Independent Variable	
Treatment	A creator's assignment to the experimental group (0 = Control, 1 = Treatment)
Creator Demographics	
Gender	A gender indicator (0 = Unknown, 1 = Male, 2 = Female)
CityLevel	The tier of a creator's located city (0 = Unknown, 1 = First-tier city, 2 = Second-tier city, 3 = Third-tier city, 4 = Fourth-tier city, 5 = Fifth-tier and over Fifth-tier city)
NumFollowers	The total number of followers that a creator has on the platform
Creator Productivity	
NumUploadVideos	The total number of videos uploaded by a creator during the experiment

ProducedVideoLength	The total length of the videos produced by a creator during the experiment (in seconds)
DaysOfProduction	The total number of days a creator produced videos during the experiment
Creator Performance	
NumLikes	The total number of likes on a creator's video content during the experiment
NumForwards	The total number of forwards on a creator's video content during the experiment
NumComments	The total number of comments on a creator's video content during the experiment
NumCompleteViews	The number of video views on a creator's content in which viewers watched at least 90% of a video's total duration during the experiment
Creator Growth	
NumNewFollowers	The total number of new followers gained by a creator during the experiment
MonetizeProbability	A binary indicator for whether a creator became eligible for platform monetization during the experiment, based on the platform's criteria (0 = Ineligible, 1 = Eligible)
Others	
NumViews	The total number of times the creator's content is presented to viewers in their feeds

Table 8: Descriptive Statistics in Study 2

Variable	***Obs.***	***Mean***	***Std***	***Min***	***Max***
Creator Demographics					
Gender	960,401	1.63	0.48	0	2
CityLevel	960,401	3.44	1.39	0	5
NumFollowers	960,401	10.31	44.92	0	999
Creator Productivity					
NumUploadVideos	960,401	3.06	17.88	0	4,690
ProducedVideoLength	960,401	1,044.11	8,490.49	0	3,297,589
DaysOfProduction	960,401	2.11	6.44	0	73
Creator Performance					
NumLikes	960,401	109.29	27,640.15	0	22,095,603
NumForwards	960,401	34.33	7,674.89	0	6,080,395
NumComments	960,401	3.68	545.02	0	340,697
NumCompleteViews	960,401	1,986.29	332,579.93	0	151,806,470
Creator Growth					
NumNewFollowers	960,401	9.58	1,571.53	-1,795	658,610
MonetizeProbability	960,401	0.02	0.12	0	1
Others					
NumViews	960,401	10,277.46	1,087,184.36	0	541,603,041

Notes: (a) To protect data confidentiality, the partner platform scaled the raw values of several variables (*NumUploadVideos, ProducedVideoLength, DaysOfProduction, NumLikes, NumForwards, NumComments, NumCompleteViews, NumNewFollowers, NumViews*). Therefore, the numerical values presented in the table represent scaled metrics rather than absolute values. (b) All variables are measured at the creator level. (c) *NumViews* is used in the Manipulation Checks in Appendix C. (d) A negative *NumNewFollowers* indicates a decrease in the number of followers. (e) *NumViews* refers to the traffic or attention directed to the creator, specifically how many times their content is presented to viewers in recommendation feeds. *NumCompleteViews* measures engagement at the creator level (the total number of times viewers watched at least 90% of the creator's videos).

4.2.3 Estimation Equation in Study 2

In Study 2, we seek to examine the impact of viewer attention allocation on long tail creators' content productivity, performance, and growth outcomes, estimating Equation (2.1) with OLS regressions:

$$OutcomeVariable_c = \beta_0 + \beta_1 \times Treatment_c + \varepsilon_c \tag{2.1}$$

The key independent variable is the treatment indicator, $Treatment_c$, which equals 1 if a long tail creator belongs to the treatment group and 0 otherwise. In terms of outcomes: for creator productivity, we examine the number of uploaded videos, produced video length, and days of production; for creator performance, we examine the number of likes, comments, forwards, and complete views; for creator growth, we examine the number of new followers and the probability of monetization.

4.2.4 Results in Study 2

We first report the impacts of viewer attention allocation on creator productivity in Table 9. The results show that long tail creators in the treatment group upload significantly more videos than those in the control group ($\hat{\beta}_1 = 0.174, p < 0.01$). In addition, treated creators significantly extend their days of production ($\hat{\beta}_1 = 0.087, p < 0.01$), indicating higher persistence and sustained content creation over time. Moreover, viewer attention allocation leads to a significant increase in creators' total video length produced ($\hat{\beta}_1 = 41.170, p < 0.05$), suggesting that increased traffic exposure encourages creators to generate a larger overall volume of content rather than merely reallocating effort across existing uploads. Taken together, these results demonstrate that viewer attention allocation operates as an effective behavioral stimulus for long tail creators, increasing their content production activities.

Table 9: Effects of Viewer Attention Allocation on Creator Productivity in Study 2

Variable	***NumUploadVideos***	***ProducedVideoLength***	***DaysOfProduction***
	(1)	***(2)***	***(3)***
Treatment	0.174***	41.170**	0.087***
	(0.036)	(17.330)	(0.013)
Observations	960,401	960,401	960,401

Notes: Robust standard errors in parentheses; $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

We then present the effects of viewer attention allocation on creator performance outcomes in Table 10. Our results show that the treatment has a positive and statistically significant effect on all

creator performance metrics, including number of likes ($\hat{\beta}_1 = 156.702$, $p < 0.01$), forwards ($\hat{\beta}_1 = 43.414$, $p < 0.01$), comments ($\hat{\beta}_1 = 3.591$, $p < 0.01$), and complete views ($\hat{\beta}_1 = 2{,}517.643$, $p < 0.01$). These results provide strong evidence that platform viewer attention allocation leads to significantly higher levels of creator performance for long tail creators' content during the experiment.

Table 10: Effects of Viewer Attention Allocation on Creator Performance in Study 2

Variable	***NumLikes***	***NumForwards***	***NumComments***	***NumCompleteViews***
	(1)	***(2)***	***(3)***	***(4)***
Treatment	156.702***	43.414***	3.591***	2,517.643***
	(56.408)	(15.663)	(1.112)	(678.730)
Observations	960,401	960,401	960,401	960,401

Notes: Robust standard errors in parentheses; $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

Next, we assess the effect of viewer attention allocation on long tail creators' growth during the field experiment in Table 11. We observe that viewer attention allocation has a positive and statistically significant effect on long tail creator growth measured by number of new followers ($\hat{\beta}_1 = 11.334$, $p < 0.01$). This result indicates that enhanced visibility effectively converts viewers into followers, expanding long tail creators' audience base. Meanwhile, the effect of viewer attention allocation on long tail creators' monetization probability is statistically insignificant across all estimation specifications (OLS, Probit, and Logit), suggesting that although attention allocation increases follower size, it does not immediately translate into a higher likelihood of monetization within the experimental period.

Table 11: Effects of Viewer Attention Allocation on Creator Growth in Study 2

Variable	***NumNewFollowers***	***MonetizeProbability***		
	(1)	***(2)***	***(3)***	***(4)***
Treatment	11.334***	0.0004	0.009	0.024
	(3.207)	(0.002)	(0.006)	(0.016)
Observations	960,401	960,401	960,401	960,401
Estimation	OLS	OLS	Probit	Logit

Notes: Robust standard errors in parentheses; $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

To conclude, allocating additional viewer attention to long tail creators substantially increases their content productivity, performance, and early-stage growth (Study 2). Meanwhile, viewer attention allocation appears to have insignificant effects on long tail creators' monetization probability during the relatively short duration of the field experiment, which is a key indicator of long-term creator success.

This raises meaningful follow-up questions: How does viewer attention allocation influence long tail creators' development? Do these developmental effects persist over time, or do they dissipate once the intervention ends? Note that the heterogeneity analyses in Study 1 reveal that active viewers are less discouraged by long tail creators' content and exhibit stronger engagement with such content than casual viewers do. Building on these insights, in Study 3, we leverage a quasi-natural experiment on Partner Platform B, and examine how viewer attention composition, namely, viewership exposure from active viewers, influences the magnitude and persistence of creators' developmental responses both short term and long term, and explore the plausible underlying mechanisms through mediation tests.

4.3 Study 3: Creator-Side Quasi-Natural Experiment

4.3.1 Quasi-Natural Experiment in Study 3

In Study 3, we leverage a quasi-natural experiment on Partner Platform B, a leading online short video platform in Asia with a substantial user base and features similar to Partner Platform A.[12] The platform implemented an exogenous viewer attention intervention for all newly registered creators during the first 30 days after creator account activation. Focusing on newly registered creators allows us to study creators at the tail of the follower distribution, which is consistent with our operational definition of long tail creators as those with fewer than 1,000 followers in Studies 1 and 2. The viewer attention intervention involves a baseline attention allocation policy that guarantees minimum content exposure of these creators. Through consultation with Partner Platform B's engineering team, we confirmed that the composition of viewer attention is determined by consistent system-level allocation rules rather than by creator-specific features or behaviors. This is consistent with prior literature documenting that recommender systems struggle to match content with appropriate audiences for new creators due to data scarcity (Chen et al., 2024; Yin et al., 2012). Consequently, this viewer attention intervention constitutes a quasi-natural experiment for long tail creators, in which variation in the composition of allocated viewership is plausibly orthogonal to creators' characteristics. To further examine the validity of this

[12] Partner Platform B allows us to observe long-term creator outcomes, which is not feasible on Partner Platform A due to a platform-wide data consolidation in late 2023 that renders creator-level performance data prior to 2024 inaccessible.

assumption, we conduct balance tests on observable creator covariates. The results of our balance checks suggest that early variation in *ActiveViewRatio* is not systematically associated with observable creator characteristics (Appendix D). Given the quasi-experimental identification, we seek to infer the effects of active viewership on long tail creators' productivity, performance, and growth, both short-term and long-term.

4.3.2 Data and Measures in Study 3

We obtained data from a random sample of 885,478 creators who registered on Partner Platform B between January 1, 2025, and January 14, 2025. For each creator, we compiled profile and behavioral data beginning from their registration date and extending through the subsequent observation period. Our independent variable, *ActiveViewRatio*, measures the proportion of views from active viewers relative to total views during each creator's first 30 days after registration. We examine effects on three sets of outcomes: creator productivity, creator performance, creator growth, both short-term and long-term. Short-term outcomes are measured over the first 30 days following each creator's registration date. These include productivity metrics (*NumUploadVideos, ProducedVideoLength, DaysOfProduction*), performance metrics (*NumLikes, NumForwards, NumComments, NumCompleteViews*), and growth metrics (*NumNewFollowers, MonetizeProbability*). Long-term outcomes aggregate the same set of variables over the 6-month period following registration.[13] We also control for creators' demographic and geographic characteristics (*Gender*, *CityLevel*, *Region*, and *CommunityType*).[14] Table 12 presents the definitions of key variables in Study 3, and Table 13 shows the descriptive statistics of these variables.

Table 12: Variable Descriptions in Study 3

Variable	***Description***
Independent Variable	
ActiveViewRatio	The proportion of views from active viewers relative to total views during each creator's first 30 days after registration
Dependent Variable	
Short-Term (the first month after registration) & Long-Term (the 6 months after registration):	

[13] For example, for a creator who registered on January 1, 2025, short-term outcomes are measured from January 1 to January 30, 2025, while long-term outcomes are measured from January 1 to June 30, 2025.
[14] These variables constitute the full set of creator-level covariates observable to the platform at the time of account registration, as these newly created accounts have no prior performance history or behavioral data.

-- Creator Productivity	
NumUploadVideos	The total number of videos uploaded by a creator
ProducedVideoLength	The total length of the videos produced by a creator (in seconds)
DaysOfProduction	The total number of days a creator produced videos
-- Creator Performance	
NumLikes	The total number of likes on a creator's video content
NumForwards	The total number of forwards on a creator's video content
NumComments	The total number of comments on a creator's video content
NumCompleteViews	The number of video views on a creator's video content in which the viewer watched at least 90% of the video's total duration
-- Creator Growth	
NumNewFollowers	The total number of new followers gained by a creator
MonetizeProbability	A binary indicator for whether a creator became eligible for platform monetization during the experiment, based on the platform's criteria (0 = Ineligible, 1 = Eligible)
Control Variable	
Gender	A gender indicator (0 = Unknown, 1 = Male, 2 = Female)
CityLevel	The tier of a creator's located city (0 = Unknown, 1 = First-tier city, 2 = Second-tier city, 3 = Third-tier city, 4 = Fourth-tier city, 5 = Fifth-tier and over Fifth-tier city)
Region	Region of the creator (0 = Unknown; 1 = Northern; 2 = Southern).
CommunityType	Community classification of the creator (0 = Unknown; 1 = Urban; 2 = Foreign; 3 = Rural; 4 = Township)

Notes: (a) We sample creators who registered between January 1, 2025, and January 14, 2025. (b) Short-term dependent variables are measured over the first 30 days following each creator's registration date (e.g., a creator registered on 2025-01-01 is observed from 2025-01-01 to 2025-01-30). (c) Long-term dependent variables are measured over the first 6 months after each creator's registration (e.g., a creator registered on 2025-01-01 is observed from 2025-01-01 to 2025-06-30).

Table 13: Descriptive Statistics in Study 3

Variable	***Obs.***	***Mean***	***Std***	***Min***	***Max***
Independent Variable					
ActiveViewRatio	885,478	0.83	0.22	0	1
Dependent Variable					
Short-Term (the first month after registration):					
-- Creator Productivity					
NumUploadVideos	885,478	11.57	108.98	0	4818
ProducedVideoLength	885,478	550.27	7,563.06	0	717269
DaysOfProduction	885,478	3.65	6.07	0	38
-- Creator Performance					
NumLikes	885,478	65.44	337.94	0	28264
NumForwards	885,478	9.27	169.59	0	93354
NumComments	885,478	19.49	150.30	0	46595
NumCompleteViews	885,478	100.28	700.58	0	54294
-- Creator Growth					
NumNewFollowers	885,478	76.71	281.27	0	44728
MonetizeProbability	885,478	0.00017	0.01	0	1

Long-Term (the 6 months after registration):					
-- Creator Productivity					
NumUploadVideos	885,478	30.21	319.29	0	20418
ProducedVideoLength	885,478	1,324.01	20,274.20	0	2085152
DaysOfProduction	885,478	9.68	21.15	0	225
-- Creator Performance					
NumLikes	885,478	225.21	1081.77	0	55505
NumForwards	885,478	31.84	408.14	0	151126
NumComments	885,478	64.86	417.91	0	75375
NumCompleteViews	885,478	261.17	1,390.45	0	97415
-- Creator Growth					
NumNewFollowers	885,478	84.62	446.20	-1249	63549
MonetizeProbability	885,478	0.00072	0.03	0	1
Control Variable					
Gender	885,478	1.29	0.62	0	2
CityLevel	885,478	2.95	1.53	0	5
Region	885,478	1.6	0.61	0	2
CommunityType	885,478	1.08	1.45	0	4

Notes: (a) To protect data confidentiality, the partner platform scaled the raw values of several variables (*NumUploadVideos, ProducedVideoLength, DaysOfProduction, NumLikes, NumForwards, NumComments, NumCompleteViews, and NumNewFollowers).* Therefore, the numerical values presented in the table represent scaled metrics rather than absolute values. (b) All variables are measured at the creator level. (c) A negative *NumNewFollowers* indicates a decrease in the number of followers.

4.3.3 Estimation Equation in Study 3

Leveraging the exogeneity of active viewer composition, we employ OLS regression following the specification in Equation (3.1). $ActiveViewRatio_c$ represents the proportion of views from active viewers for creator c during the creator's first month after registration (ranging from 0 to 1). $OutcomeVariable_c$ denotes the outcome variables of interest, including both short-term outcomes measured over the first 30 days following registration and long-term outcomes aggregated over the first 6 months following registration. $Controls_c$ includes the full set of demographic and geographic control variables, namely gender, city level, region, and community type, to account for baseline heterogeneity across creators. The coefficient β_1 captures the effect of early active-view exposure on creator outcomes. We further conduct robustness checks by re-estimating the models without control variables (Appendix E) and with log transformations (Appendix F) and find that the results remain consistent.

$$OutcomeVariable_c = \beta_0 + \beta_1 \times ActiveViewRatio_c + \beta_2 \times Controls_c + \varepsilon_c, \quad (3.1)$$

4.3.4 Results in Study 3

First, we examine whether *ActiveViewRatio* (the proportion of views from active viewers relative to total views) affects creators' productivity during the first month following registration. Table 14 reports the estimated effects across three productivity-related outcomes, indicating that higher early active-view exposure is associated with significantly higher creator productivity. Specifically, *ActiveViewRatio* has positive and statistically significant effects on the number of videos uploaded ($\hat{\beta}_1 = 11.028, p < 0.01$), total video length ($\hat{\beta}_1 = 589.146, p < 0.01$), and days of production ($\hat{\beta}_1 = 1.258, p < 0.01$). Additionally, we analyze how *ActiveViewRatio* affects creators' performance in the short run. Table 15 reports the estimated effects across four performance metrics. The results show that a higher share of views generated by active viewers leads to significantly stronger short-term creator performance. Specifically, *ActiveViewRatio* has positive and statistically significant effects on the number of likes ($\hat{\beta}_1 = 22.848, p < 0.01$), forwards ($\hat{\beta}_1 = 1.258, p < 0.01$), comments ($\hat{\beta}_1 = 6.035, p < 0.01$), and complete views ($\hat{\beta}_1 = 88.627, p < 0.01$) gained by a long tail creator. Further, we examine the effect of *ActiveViewRatio* on creators' growth outcomes during the first month following registration. Table 16 shows the estimated effects on follower growth and monetization. The results show that higher early active-view exposure is associated with significantly stronger creator growth during the first month following registration. Specifically, *ActiveViewRatio* has a positive and statistically significant effect on the number of new followers ($\hat{\beta}_1 = 14.092, p < 0.01$) and increases the probability of monetization ($\hat{\beta}_1 = 0.0003, p < 0.01$). The results on monetization are also robust with nonlinear specifications, with positive and highly significant effects in both the Probit and Logit models. Overall, these short-term results largely align with Study 2 and further reveal that higher active-viewer exposure drives stronger short-term outcomes for long tail creators, enhancing their productivity, performance, and growth during the first month following registration.

Table 14: Short-term Effects on Creator Productivity in Study 3

Variable	*NumUploadVideos* (1)	*ProducedVideoLength* (2)	*DaysOfProduction* (3)
	11.028***	589.146***	1.258***

Variable	*NumUploadVideos*	*ProducedVideoLength*	*DaysOfProduction*
	(1)	*(2)*	*(3)*
ActiveViewRatio	(0.413)	(25.655)	(0.031)
Controls	Yes	Yes	Yes
Observations	885,478	885,478	885,478

Notes: (a) Partner Platform B scaled the raw values of outcome variables in Study 3. Therefore, the coefficients presented in this and subsequent tables represent scaled metrics rather than absolute values. (b) Robust standard errors in parentheses; $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

Table 15: Short-term Effects on Creator Performance in Study 3

Variable	*NumLikes*	*NumForwards*	*NumComments*	*NumCompleteViews*
	(1)	*(2)*	*(3)*	*(4)*
ActiveViewRatio	22.848***	1.258***	6.035***	88.627***
	(0.970)	(0.385)	(0.371)	(1.927)
Controls	Yes	Yes	Yes	Yes
Observations	885,478	885,478	885,478	885,478

Notes: Robust standard errors in parentheses; $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

Table 16: Short-term Effects on Creator Growth in Study 3

Variable	*NumNewFollowers*	*MonetizeProbability*		
	(1)	*(2)*	*(3)*	*(4)*
ActiveViewRatio	14.092***	0.0003***	0.698***	2.579***
	(0.719)	(0.0001)	(0.159)	(0.615)
Controls	Yes	Yes	Yes	Yes
Observations	885,478	885,478	885,478	885,478
Estimation	OLS	OLS	Probit	Logit

Notes: Robust standard errors in parentheses; $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

Next, we investigate the long-term outcomes, starting with testing the influence of *ActiveViewRatio* on creator productivity over the 6-month period following registration, with results reported in Table 17. The findings indicate that higher early active-view exposure is associated with significantly higher creator productivity in the long run. Specifically, *ActiveViewRatio* has positive and statistically significant effects on the number of videos uploaded ($\hat{\beta}_1 = 30.505, p < 0.01$), total video length ($\hat{\beta}_1 = 1573.230, p < 0.01$), and days of production ($\hat{\beta}_1 = 4.480, p < 0.01$). We then examine the effects of *ActiveViewRatio* on creator performance over the 6 months, with results reported in Table 18. Our results show that *ActiveViewRatio* has positive and statistically significant effects on the number of likes ($\hat{\beta}_1 = 81.161, p < 0.01$), forwards ($\hat{\beta}_1 = 5.921, p < 0.01$), comments ($\hat{\beta}_1 = 23.971, p < 0.01$), and complete views ($\hat{\beta}_1 = 188.238, p < 0.01$). Lastly, we examine the long-term effects of *ActiveViewRatio* on creator growth, with results reported in Table 19. The findings indicate that the

positive effects of early active-view exposure persist in the long run for both follower accumulation and monetization outcomes. Specifically, *ActiveViewRatio* has a positive and statistically significant effect on the number of new followers ($\hat{\beta}_1 = 25.640, p < 0.01$), and monetization probability ($\hat{\beta}_1 = 0.001, p < 0.01$).

To further examine the dynamic evolution of the long-term effects, we utilize monthly data following creators' registration and estimate separate regressions of each outcome on *ActiveViewRatio* for every post-registration month. We then plot the estimated coefficients along with their confidence intervals to illustrate how the effect of early active-view exposure evolves over time. We report our long-term dynamic effect plots in Appendix G, showing relatively persistent long-term effects. In sum, our evidence suggests that early higher active-view exposure substantially improves long tail creators' performance, growth, and productivity in both the short and long term.

Table 17: Long-term Effects on Creator Productivity in Study 3

Variable	***NumUploadVideos***	***ProducedVideoLength***	***DaysOfProduction***
	(1)	***(2)***	***(3)***
ActiveViewRatio	30.505***	1573.230***	4.480***
	(1.240)	(70.983)	(0.087)
Controls	Yes	Yes	Yes
Observations	885,478	885,478	885,478

Notes: Robust standard errors in parentheses; $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

Table 18: Long-term Effects on Creator Performance in Study 3

Variable	***NumLikes***	***NumForwards***	***NumComments***	***NumCompleteViews***
	(1)	***(2)***	***(3)***	***(4)***
ActiveViewRatio	81.161***	5.921***	23.971***	188.238***
	(3.091)	(0.993)	(1.157)	(4.668)
Controls	Yes	Yes	Yes	Yes
Observations	885,478	885,478	885,478	885,478

Notes: Robust standard errors in parentheses; $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

Table 19: Long-term Effects on Creator Growth in Study 3

Variable	***NumNewFollowers***	***MonetizeProbability***		
	(1)	***(2)***	***(3)***	***(4)***
ActiveViewRatio	25.640***	0.001***	0.346***	1.168***
	(1.364)	(0.0001)	(0.064)	(0.228)
Controls	Yes	Yes	Yes	Yes
Observations	885,478	885,478	885,478	885,478
Estimation	OLS	OLS	Probit	Logit

Notes: Robust standard errors in parentheses; $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

4.3.5 Mediation Analyses in Study 3

Having established that a higher proportion of views from active viewers (relative to total views) has significant positive effects on long tail creators' productivity, performance, and growth in the long run, we further explore the plausible underlying mechanisms. Drawing on prior literature (Oestreicher-Singer & Zalmanson, 2013), viewer attention could lead to a) passive content viewership and b) active viewer engagement behaviors (e.g., likes, comments, forwards, complete views), which manifest as engagement performance metrics for creators. And active viewers are more likely to exhibit active content engagement (Heinonen, 2011). Bearing the above in mind, we posit that active viewers are more likely to actively engage with long tail creators' content, which could create positive, engagement-mediated mechanisms: higher initial active-view exposure leads to greater engagement performance for the creators (likes, forwards, comments, and complete views), which in turn stimulates increased long-term creator productivity, follower growth and monetization potential.

We then conduct mediation analyses to examine whether early engagement performance serves as plausible mechanism linking initial viewer composition to long tail creators' subsequent productivity and growth. Using engagement performance metrics aggregated over months 2-3 as mediators and creator productivity and growth outcomes in months 4-6 as dependent variables, we estimate mediation effects using PROCESS Model 4 (Hayes, 2022) with 5,000 bootstrap resamples on a random subsample of 100,000 long tail creators. The mediation results are presented in Table 20. To begin with, across the productivity outcomes, our results show consistent evidence of engagement-based mediation effects. For example, the engagement performance metrics (e.g., likes, comments, forwards, complete views) appear to be significant mediators between *ActiveViewRatio* and the long tail creators' productivity indicators (e.g., number of uploaded videos and days of production). Also, for the outcome of produced total video length, the significant mediators are primarily complete views and comments. Further, for long tail creators' growth outcomes, *ActiveViewRatio* operates primarily through forwards as the key mechanism towards new follower acquisition. Meanwhile, the number of likes, comments, and complete views are

significant mediators between *ActiveViewRatio* and the long tail creators' monetization probability. Taken together, our findings provide convergent evidence for engagement-based mechanisms (though the dominant engagement channel varies by outcome): Long tail creators' early exposure to a higher share of active viewers increases subsequent viewer engagement performance, which in turn helps cultivate the long-term productivity and development of those creators.

Table 20: Mediation Analysis of Engagement Performance on Creator Productivity and Growth

	Average Effect	95% CI
DV = *NumUploadVideos* (Months 4–6)		
Mediation (Total Indirect Effect)	0.604	[0.458, 0.784]
-- Likes (Months 2–3)	0.111	[0.005, 0.233]
-- Forwards (Months 2–3)	0.014	[0.004, 0.033]
-- Comments (Months 2–3)	0.125	[0.051, 0.212]
-- Complete Views (Months 2–3)	0.353	[0.181, 0.588]
Direct Effect	4.292	[1.914, 6.670]
Total Effect	4.896	[2.512, 7.281]
DV = *ProducedVideoLength* (Months 4–6)		
Mediation (Total Indirect Effect)	17.894	[11.456, 26.645]
-- Likes (Months 2–3)	-2.941	[-9.177, 1.566]
-- Forwards (Months 2–3)	0.005	[-0.461, 0.552]
-- Comments (Months 2–3)	5.417	[0.064, 15.242]
-- Complete Views (Months 2–3)	15.413	[7.834, 25.831]
Direct Effect	286.039	[69.601, 502.478]
Total Effect	303.933	[87.456, 520.411]
DV = *DaysOfProduction* (Months 4–6)		
Mediation (Total Indirect Effect)	0.195	[0.158, 0.237]
-- Likes (Months 2–3)	0.070	[0.041, 0.103]
-- Forwards (Months 2–3)	0.009	[0.003, 0.018]
-- Comments (Months 2–3)	0.082	[0.053, 0.122]
-- Complete Views (Months 2–3)	0.033	[0.020, 0.049]
Direct Effect	0.765	[0.553, 0.976]
Total Effect	0.961	[0.740, 1.181]
DV = *NumNewFollowers* (Months 4–6)		
Mediation (Total Indirect Effect)	0.426	[0.051, 0.847]
-- Likes (Months 2–3)	0.158	[-0.304, 0.699]
-- Forwards (Months 2–3)	0.231	[0.052, 0.471]
-- Comments (Months 2–3)	0.088	[-0.318, 0.551]
-- Complete Views (Months 2–3)	-0.052	[-0.261, 0.141]
Direct Effect	-3.100	[-8.306, 2.106]
Total Effect	-2.673	[-7.889, 2.542]
DV = *MonetizeProbability* (Months 4–6)		
Mediation (Total Indirect Effect)	0.019	[0.015, 0.024]
-- Likes (Months 2–3)	0.008	[0.005, 0.014]
-- Forwards (Months 2–3)	0.001	[-0.001, 0.001]
-- Comments (Months 2–3)	0.005	[0.002, 0.010]
-- Complete Views (Months 2–3)	0.006	[0.003, 0.008]

Direct Effect	1.038	[-0.579, 2.654]
Total Effect	—	—

Notes: (a) Independent variable = *ActiveViewRatio*. Mediators = *NumLikes, NumForwards, NumComments,* and *NumCompleteViews* measured over Months 2–3. Dependent variables = *NumUploadVideos, ProducedVideoLength, DaysOfProduction*, *NumNewFollowers*, *MonetizeProbability* measured over Months 4–6. (b) Mediation analyses are conducted using PROCESS Model 4 (Hayes, 2022). Due to computational constraints, mediation analyses are performed on a random subsample of 100,000 creators. (c) Reported effects include the total indirect effect, specific indirect effects, direct effect, and total effect. (d) *MonetizeProbability* is a binary outcome. For this outcome, PROCESS estimates mediation models using logistic regression and reports effects in log-odds; the total effect is not reported by PROCESS for binary outcomes. (e) Confidence intervals in square brackets are 95% percentile bootstrap confidence intervals based on 5,000 bootstrap resamples.

5 Discussions

5.1 Implications for Research

This work offers several important implications for research. Our findings extend prior literature on the long tail theory (Anderson, 2006; Brynjolfsson et al., 2010) from retail to short video platforms. The long tail theory was originally developed in online retail contexts where variety expansion increased consumer surplus and was largely costless (Anderson, 2006; Brynjolfsson et al., 2010). In short video platforms, we show that the long tail is not passive inventory but an active claimant on scarce viewer attention. Across three complementary studies, we provide rigorous evidence that exposure to long tail content can reduce short-term viewer engagement, yet targeted attention allocation to long tail creators catalyzes their development, persistence, and long-term economic viability. Additionally, our heterogeneity analysis by viewer heterogeneity shows that the welfare impact of long tail exposure depends on who consumes it, as active viewers demonstrate greater engagement with long tail creators' content. Our findings also advance the long tail theory by pointing to the potential of boundary conditions that influence whether and when long tail offerings generate value or impose costs.

Next, to our knowledge, we are among the first to systematically investigate how platforms can effectively develop long tail creators who face persistent cold-start and discovery challenges (Lika et al., 2014). Prior research has established that star performers generate disproportionate value (Han & Ravid, 2020; Maclean, 2021) and that their exit can create substantial negative shocks (Forderer et al., 2025; Qian & Xie, 2025), yet comparatively little is known about how platforms can effectively manage and develop the long tail of creators who face persistent cold-start and discovery challenges (Lika et al.,

2014). Long tail creators are central to platforms' future growth since they supply the inflow of new content and represent the pool from which future star creators emerge. Meanwhile, recommender systems on short video platforms generally gravitate toward established creators with proven engagement and anticipated revenue from advertising and sponsorships, which creates a tension: Continually prioritizing star creators optimizes current profits but gradually weakens the creator ecosystem by discouraging long tail creators, narrowing the talent pipeline, and concentrating power among a few creators. Our work demonstrates that algorithmic attention allocation can effectively stimulate long tail creators' productivity, performance, and growth, which extends prior work on platform interventions for creator development. Further, this work adds to the growing body of work on algorithmic management (Kellogg et al., 2020; Kadolkar et al., 2025), by advancing the consideration of algorithmic management from short-term operational control, coordination, and monitoring to long-term human capital development. For example, on gig economy platforms, algorithmic management primarily affects task assignment and worker evaluation within relatively short time horizons (Duggan et al., 2023). Our results provide evidence that algorithms can also function as strategic tools for long tail creator cultivation on short video platforms. For long tail creators who struggle with initial visibility in crowded digital markets, algorithmic allocation of viewer attention reduces barriers to follower building and provides opportunities for performance signaling that would otherwise be unavailable.

5.2 Implications for Practice

Our work offers useful practical implications for platform operators and content creators. Our results indicate that platform operators face a trade-off between short-term viewer engagement and long-term ecosystem health. While our Study 1 reveals that long tail creators' content imposes short-term engagement costs, platforms need to balance these immediate costs against the strategic necessity of nurturing potential future stars and reducing dependence on established creators. Importantly, our heterogeneity analysis reveals that active viewers demonstrate greater engagement with long tail creators' content, suggesting a pathway to mitigate this trade-off. Specifically, platforms should prioritize allocating traffic from active viewer segments to long tail creators, as these viewers provide more

meaningful engagement signals that stimulate creator productivity and follower growth. Our Study 3 demonstrates that the initial exposure period is particularly helpful, as early active-view attention allocation generates sustained positive effects on creator development over time. This suggests that viewer attention allocation represents an effective algorithmic intervention that can improve not only creator productivity but also performance, growth, and monetization outcomes. For long tail creators, our findings suggest the importance of strategically acquiring high engagement viewer attention in the early stages. When platforms offer temporal boosts to content visibility in recommender systems, creators want to improve not only the quantity of exposure but also viewer engagement. Creators should focus on attracting viewers who provide active engagement through likes, comments, and complete views, as these engagement traces could serve as training data for the recommender system, improving the algorithm's ability to assess long tail creators' potential and route future viewer attention.

5.3 Limitations and Future Research

This work has several limitations that also present opportunities for future research. First, the interventions in our field experiments (Studies 1 and 2) were not sustained over an extended period, as consistently removing long tail creators or allocating traffic to them could be very harmful to platform operations and user experience. Future studies, if conditions permit, could conduct field experiments with larger observation windows to explore the long-term effects of attention allocation strategies. Second, our research context is online short video platforms, which are characterized by fast feedback loops, limited audience attention spans, and algorithm-driven discovery. These characteristics may differ from other content platforms such as music streaming services, podcasting, or long-form video platforms, where content consumption patterns may vary substantially due to user-driven discovery mechanisms. Future research should examine whether our findings generalize across different digital content ecosystems. Additionally, our setting focuses on largely independent long tail creators, whereas organizational content producers with internal analytics and resources may rely less on platform-mediated feedback for learning. Also, our results speak to the variation in engagement costs of long tail content exposure by viewer heterogeneity. If active viewers implicitly (or explicitly) subsidize platform ecosystem development by

serving as 'training audiences' for the long tail, future studies could investigate whether such costs affect viewers over time, and how platforms might design compensatory mechanisms or disclosure practices to sustain viewer engagement, satisfaction, and retention. Third, in Study 3, we leveraged a quasi-natural experiment utilizing the first 30 days after creator registration, during which the platform provided exogenous attention allocation to newly registered creators. As the treatment (active view ratio) was not randomized in a field experiment setting, we do not infer causality in our findings, although we believe endogeneity concerns are relatively minimal in a large-scale platform ecosystem. Future research could employ fully randomized experimental designs to further address these concerns. We hope our work will inspire future research to overcome these limitations in advancing our understanding of long tail theory, attention allocation strategies, and digital content creation.

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